

Tail-universal scalar sets for nonunit adjoint eigenvalues

Candidate partial result for arXiv:1509.04912

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Status. Candidate substantial partial result, likely valid; human review recommended. The theorem gives a uniform positive answer under an explicit scalar-set condition in the nonunit-eigenvalue regime. It does not classify all scalar sets and therefore does not fully solve Question 3.

1 Source problem

The source is S. Charpentier, R. Ernst, and Q. Menet, Γ -supercyclicity, arXiv:1509.04912 [1]. For a bounded operator T on a complex Banach space X , a vector x is Γ -supercyclic when

$$\text{Orb}(\Gamma x, T) = \{\gamma T^n x : \gamma \in \Gamma, n \geq 0\}$$

is dense in X . Question 3 asks which $\Gamma \subset \mathbb{C}$ make supercyclicity equivalent to Γ -supercyclicity.

Theorem (Shkarin Theorem). *Let X be a complex Banach space and $T \in L(X)$. T is $\mathbb{R}^+$ -supercyclic if and only if T is supercyclic and either the point spectrum $\sigma_p(T^*)$ is empty or $\sigma_p(T^*) = \{re^{i\theta}\}$, with $r \neq 0$ and $\theta \in \pi(\mathbb{R} \setminus \mathbb{Q})$.*

More generally the following question arises, an answer to which would involve the spectrum of T^* .

Question 3. For which $\Gamma \subset \mathbb{C}$ is supercyclicity equivalent to Γ -supercyclicity?

1.2. Statements of the main results. The purpose of this article is to discuss Questions 1, 2 and 3. In order to deal with Questions 1 and 2 we introduce two properties that a subset of $\mathbb{C}$ can enjoy or not.

Figure 1: Question 3 on page 3 of the source paper.

The source completely treats adjoint eigenvalues of modulus one in Theorem C(1), gives a warning example for nonunit eigenvalues in Theorem C(2), and then explicitly says that the nonunit and empty-spectrum cases remain unclear.

To avoid a convention-dependent conjugation, the result below is stated using an *eigenfunctional multiplier*: $f \circ T = \mu f$. Under the complex-linear dual convention this is exactly $T^*f = \mu f$; under the conjugate-dual convention used in the source, the corresponding adjoint eigenvalue is $\bar{\mu}$. In either convention the modulus is unchanged.

Furthermore we give a general answer to Question 3 when $\sigma_p(T^*)$ is non-empty.

Theorem C. *Let X be a complex Banach space.*

- (1) *For every $\theta \in \mathbb{R}$, $\Gamma \subset \mathbb{C}$ satisfies the property: "For every $T \in L(X)$ with $\sigma_p(T^*) = \{e^{i\theta}\}$ and every $x \in X$*

x is Γ -supercyclic for T if and only if x is supercyclic for T "

if and only if ΓG_θ is dense in $\mathbb{C}$, where G_θ stands for the subgroup of $\mathbb{T}$ generated by $e^{i\theta}$.

- (2) *For any $r \neq 1$ and any $\theta \in \mathbb{R}$, there exist $T \in L(X)$ with $\sigma_p(T^*) = \{re^{i\theta}\}$ and $\Gamma \subset \mathbb{C}$ satisfying $\overline{\Gamma G_\theta} = \mathbb{C}$, such that T is supercyclic but not Γ -supercyclic.*

Remark 1.3. (1) For every $\theta \in \mathbb{R}$, Theorem C provides a complete understanding of the problem of describing those subsets Γ of $\mathbb{C}$ such that for every X , every $T \in L(X)$ with $\sigma_p(T^*) = \{e^{i\theta}\}$ and every $x \in X$,

x is Γ -supercyclic for T if and only if x is supercyclic for T .

Nevertheless such a problem for every T (namely for those T such that $\sigma_p(T^*) = \emptyset$ or $\sigma_p(T^*) = \{re^{i\theta}\}$ with $r \neq 1$) remains unclear.

- (2) One shall notice that, when $\theta \in \pi(\mathbb{R} \setminus \mathbb{Q})$, the condition $\overline{\Gamma G_\theta} = \mathbb{C}$ is equivalent to $\overline{\Gamma \mathbb{T}} = \mathbb{C}$.
(3) It is worth observing that the equivalence between Γ -supercyclicity and supercyclicity for T with $\sigma_p(T^*) = re^{i\theta}$, $\theta \in \pi(\mathbb{R} \setminus \mathbb{Q})$, depends on r . This differs from the particular case of the $\mathbb{R}_+$ -supercyclicity which is treated in Shkarin Theorem.

Figure 2: Theorem C and the unresolved nonunit case on page 4 of the source.

2 Proof intuition

Suppose $f \circ T = \mu f$. A projective approximation $\alpha_k T^{n_k} x \rightarrow y$ is visible through f as $\alpha_k \mu^{n_k} f(x) \rightarrow f(y)$. Thus, for targets outside the hyperplane $\ker f$, the correct projective coefficient is asymptotically determined after multiplication by μ^{n_k} . If the rescaled sets $\mu^n \Gamma$ approach every scalar at every sufficiently late time, one can replace α_k by coefficients from Γ without changing the vector limit. The source's phase-group condition does not provide this time synchronization when $|\mu| \neq 1$; tail universality does.

The condition is nonvacuous and does not merely say that Γ is dense. Place a finer and larger finite lattice F_n at scale μ^{-n} . After multiplication by μ^n the n th block becomes F_n , while in the original scale the blocks either collapse to zero or escape to infinity when $|\mu| \neq 1$.

3 Tail-universal scalar sets

Definition 1. Let $\mu \in \mathbb{C} \setminus \{0\}$. A set $\Gamma \subset \mathbb{C}$ is *tail-universal* for μ if

$$\forall c \in \mathbb{C}, \quad \text{dist}(c, \mu^n \Gamma) \longrightarrow 0 \quad (n \rightarrow \infty). \quad (1)$$

Lemma 2 (Projective tails). *If x is a supercyclic vector for T , then for every $N \geq 0$,*

$$\bigcup_{n \geq N} \mathbb{C} T^n x$$

is dense in X .

Proof. The one-dimensional case is immediate. Assume $\dim X > 1$. If some nonempty open set U missed the displayed tail, then density of the full projective orbit would imply that every nonempty open subset of U meets the finite union $\bigcup_{n < N} \mathbb{C} T^n x$. Hence this finite union of closed one-dimensional subspaces would be dense in U , and therefore would contain U . A finite union of proper linear subspaces has empty interior, a contradiction. $\square$

Theorem 3 (Uniform spectral transfer). *Let X be a complex Banach space, $T \in \mathcal{L}(X)$, and suppose that $f \circ T = \mu f$ for some $f \in X^* \setminus \{0\}$ and $\mu \neq 0$. If Γ is tail-universal for μ , then, for every $x \in X$,*

$$x \text{ is supercyclic for } T \iff x \text{ is } \Gamma\text{-supercyclic for } T.$$

Proof. The reverse implication is immediate because $\Gamma \subset \mathbb{C}$. Suppose that x is supercyclic. Then $f(x) \neq 0$, since otherwise $f(T^n x) = \mu^n f(x) = 0$ for every n and the projective orbit would lie in the proper closed hyperplane $\ker f$.

Fix $y \in X$ with $f(y) \neq 0$. By Lemma 2, there are integers $n_k \geq k$ and scalars α_k such that

$$\alpha_k T^{n_k} x \longrightarrow y.$$

Applying f and dividing by $f(x)$ gives

$$\alpha_k \mu^{n_k} \longrightarrow c := \frac{f(y)}{f(x)} \neq 0.$$

By (1), choose $\gamma_k \in \Gamma$ with $\gamma_k \mu^{n_k} \rightarrow c$. Since both scaled coefficient sequences converge to the same nonzero number,

$$\frac{\gamma_k}{\alpha_k} = \frac{\gamma_k \mu^{n_k}}{\alpha_k \mu^{n_k}} \longrightarrow 1.$$

Consequently $\gamma_k T^{n_k} x \rightarrow y$. Thus the closure of $\text{Orb}(\Gamma x, T)$ contains $X \setminus \ker f$, which is dense in X . $\square$

4 Countable non-dense examples

Proposition 4. *For every $\mu \in \mathbb{C} \setminus \{0\}$ with $|\mu| \neq 1$, there is a countable non-dense set $\Gamma_\mu \subset \mathbb{C}$ that is tail-universal for μ . One explicit choice is*

$$F_n = \left\{ \frac{j + ik}{n} : j, k \in \mathbb{Z}, j^2 + k^2 \leq n^4 \right\}, \quad \Gamma_\mu = \bigcup_{n \geq 1} \mu^{-n} F_n.$$

Proof. Fix $c \in \mathbb{C}$. Round $n\Re c$ and $n\Im c$ to nearest integers j_n and k_n . For all sufficiently large n , $j_n^2 + k_n^2 \leq n^4$, so $q_n = (j_n + ik_n)/n$ belongs to F_n , and

$$|q_n - c| \leq \frac{\sqrt{2}}{2n}.$$

Because $F_n \subset \mu^n \Gamma_\mu$, condition (1) follows.

It remains to prove non-density. Write $r = |\mu|$. If $r > 1$, the n th block $\mu^{-n} F_n$ is contained in the disk of radius nr^{-n} , which tends to zero. Apart from the common point zero, only finitely many blocks can meet any fixed annulus bounded away from zero, so the union is not dense. If $0 < r < 1$, every nonzero point of the n th block has modulus at least r^{-n}/n , which tends to infinity. Hence every bounded disk contains nonzero points from only finitely many finite blocks, and again the union is not dense. Countability is immediate. $\square$

Remark 5. Combining Theorem 3 and Proposition 4 gives, for each nonunit μ , a single countable non-dense scalar set that works simultaneously for every supercyclic operator possessing the adjoint spectral multiplier μ and for every one of its supercyclic vectors. This is stronger than constructing a set for one selected operator.

5 Verification, novelty, and limitations

The proof is analytic. The accompanying checker verifies finite lattice rounding errors and the two block-scale asymptotics for representative eigenvalues on either side of the unit circle. These computations are sanity checks only.

A bounded search on 9 August 2026 covered:

- the run indexes and the source paper;
- arXiv:1711.10932, arXiv:2005.11230, and arXiv:2411.03179;
- exact or close searches for Γ -supercyclicity, nonunit adjoint eigenvalue, $\mu^n \Gamma$, and scaled scalar sets.

No explicit version of Theorem 3 or Proposition 4 was found. Novelty confidence is moderate.

The result is partial: condition (1) is sufficient, but no necessity theorem is claimed; the source's empty adjoint point-spectrum case also remains open. Human review should focus on Lemma 2, the coefficient-ratio step in Theorem 3, the non-density proof, and the bounded novelty assessment.

References

- [1] S. Charpentier, R. Ernst, and Q. Menet, Γ -supercyclicity, *Integral Equations Operator Theory* 85 (2016), 71–92; arXiv:1509.04912.